



Fig. 7: Christmas tree lighting

provide supply to melody generator IC9 (UM66) and transistor T2. The output from UM66 drives the speaker with the help of transistor T2.

Fig. 2 shows the circuit of power supply. The 230V AC mains is stepped down by transformer

X1 to deliver the secondary output of 9V, 1.5A. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C5 and then regulated by IC 78S05 (IC10). Capacitor C6 bypasses the ripples present in the regulated 5V power supply. Use a proper heat-sink for IC 78S05.

Construction

An actual-size, single-side PCB for the circuit within the dotted lines is shown in Fig. 3 and its component layout in Fig. 4. CON2 through CON8 are used for extension of IC3 through IC8, respectively. Set the 1Hz frequency by varying preset VR1. LED1 glows with this frequency. LED2 through LED5 glow in binary pattern. The musical tone plays when pin 13 of IC2 goes high.

The circuit around IC 74HC164 is used for extension to avoid running more wires in order to connect the LEDs. You can use this circuit at a suitable point for decoration.

Assemble the circuit wired around IC 74HC164 (IC3) on a PCB. An actual-size, single-side PCB for this circuit is shown in Fig. 5 and its component layout in Fig. 6. Suitable connectors are provided for extension. Separate PCB (Fig. 5) is used for extension of single stage of 74HC164. There is a lot of scope to vary the display pattern by using LEDs of different sizes, shapes and colours. Another interesting variation is to try a tri-colour LED.

The circuit can be mounted as desired for decoration. For example, you can fasten flashing LEDs to a Christmas tree. Another idea is to cut out a cardboard in the shape of a star, decorate it with tinsel and mount LEDs on the points of the star. The LEDs can be easily fixed in position with fine wires taken back to the circuit board. When hung on a Christmas tree, the decoration light looks as shown in Fig. 7. ●

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THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT



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